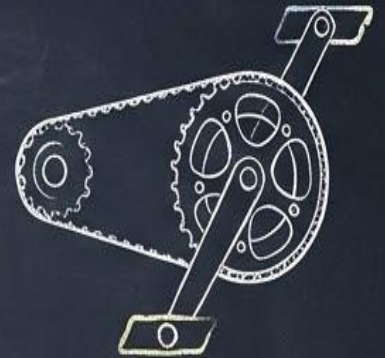
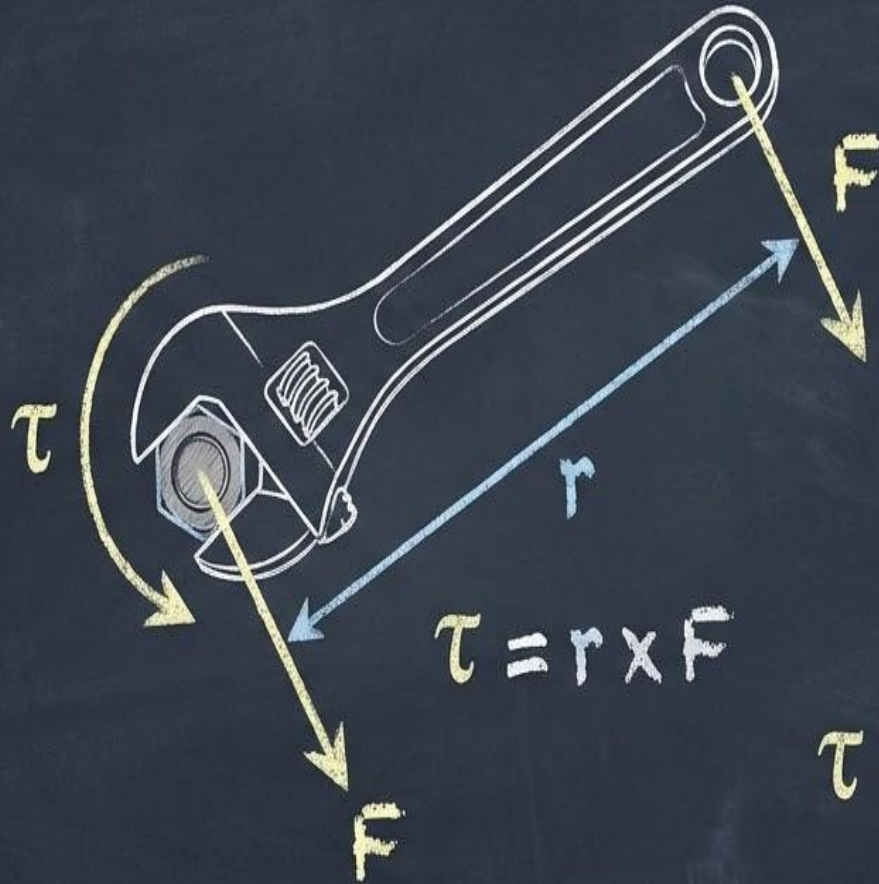


Rotational Statics i.e. “Torque”

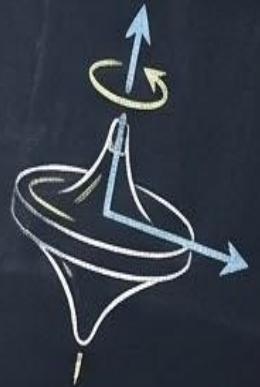
LET'S TORQUE ABOUT IT!



$$\tau = rF \sin \theta$$

$$\tau = \mathbf{r} \times \mathbf{F}$$

$$\tau = I\alpha$$



The rotational equivalent of linear force.

Net force vs. net torque

- Greater net force causes greater linear acceleration.
- Greater net torques cause greater rotational or angular accelerations.

Torque is a twist or turn that tends to produce rotation. * *

* Applications are found in many common tools around the home or industry where it is necessary to turn, tighten or loosen devices.



Definition of Torque

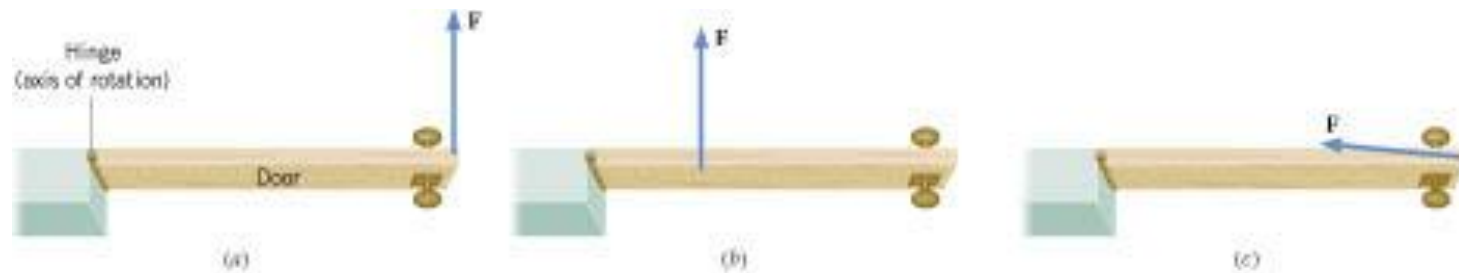
Torque is defined as the tendency to produce a change in rotational motion.

Examples:



Torque

- Torque depends on
 - the magnitude of the force
 - the point where the force is applied relative to the axis of rotation
 - the direction of the force



$$\tau = F \cdot r$$

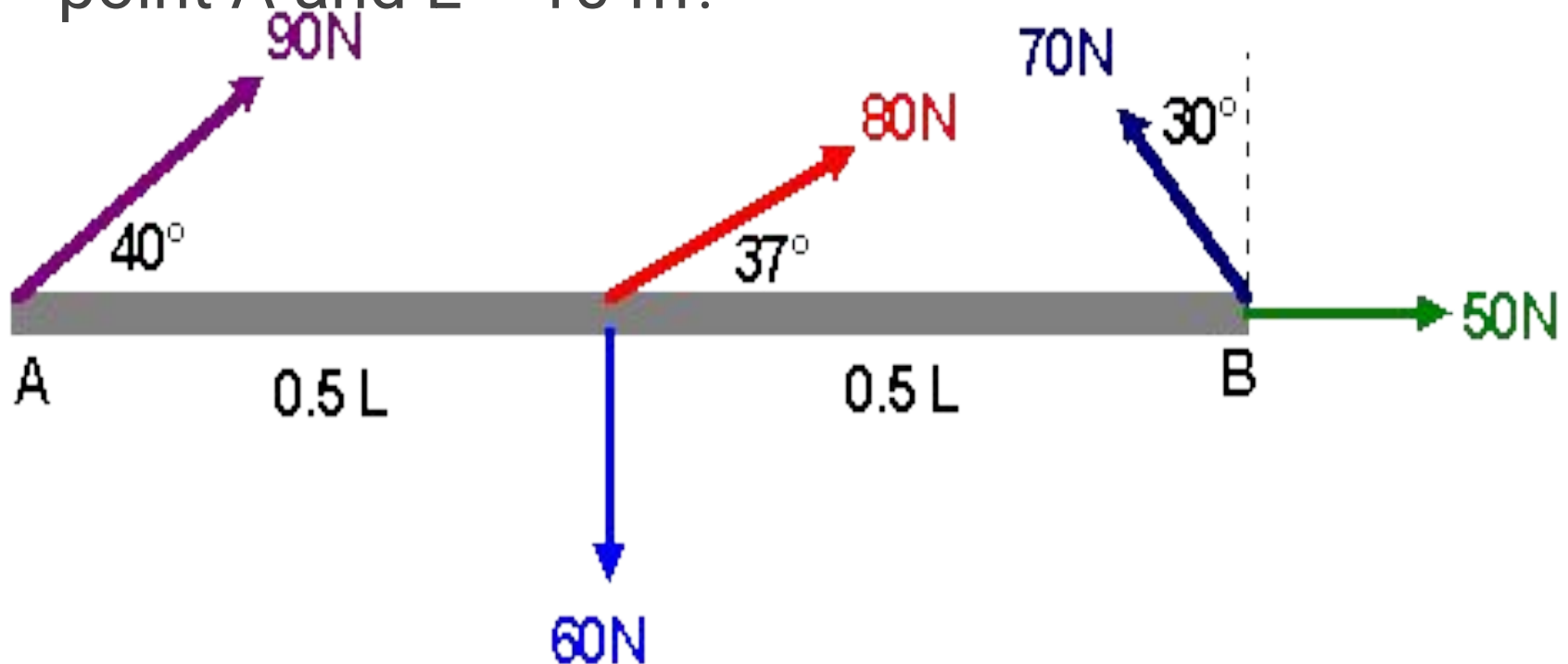
Calculating Torque

$$\tau = r \times F$$

- Torque = (lever arm) x (magnitude of force)
- r and F must be **PERPENDICULAR**
- Unit = N·m
- Direction
 - Positive torque causes counterclockwise rotation
 - Negative torque causes clockwise rotation

Sample Problem

- In the diagram below, calculate the torque exerted by each force if the rod has a pivot at point A and $L = 10\text{ m}$?



Equilibrium

- For a rigid body to be in equilibrium, two conditions must be met.
 - Linear motion cannot change
 - therefore $\Sigma F_x = 0$ and $\Sigma F_y = 0$
 - Rotational motion cannot change
 - therefore $\Sigma \tau = 0$
- “A rigid body is in equilibrium if it has zero translational acceleration and zero rotational acceleration.”

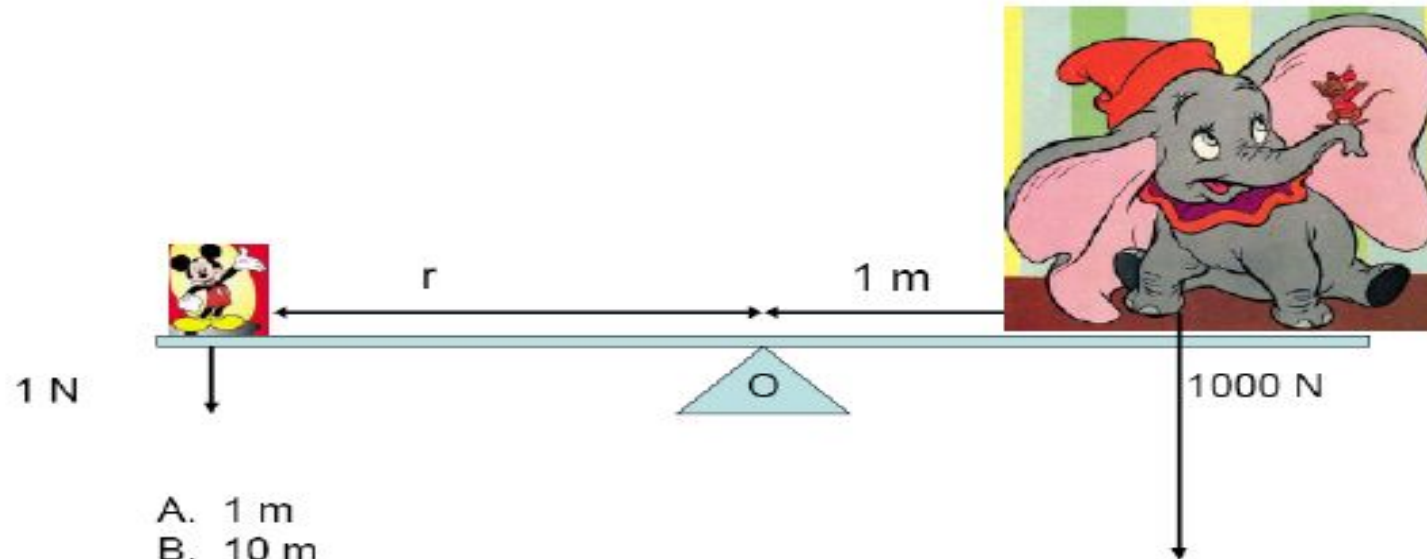
Center of gravity

- The center of gravity (cg) of a rigid body is the point at which its weight can be considered to act when calculating the torque due to the weight.
 - For a symmetric shape the cg is located at the geometric center.

Example



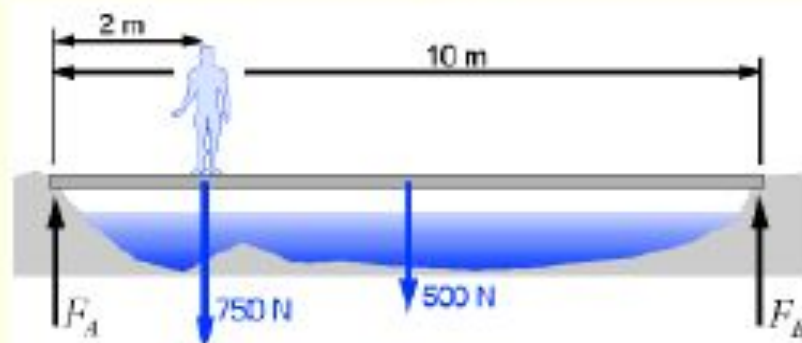
How long should the seasaw be for Mickey to balance Dumbo ?



- A. 1 m
- B. 10 m
- C. 100 m
- D. 1000 m

Example

- For example, consider a 10-meter bridge that weighs 500 newtons supported at both ends. A person who weighs 750 newtons is standing 2 meters from one end of the bridge.
- What are the forces (F_A , F_B) holding the bridge up at either end?



Practice

43. **III** A 3.0-m-long rigid beam with a mass of 100 kg is supported at each end, as shown in **Figure P8.43**. An 80 kg student stands 2.0 m from support 1. How much upward force does each support exert on the beam?

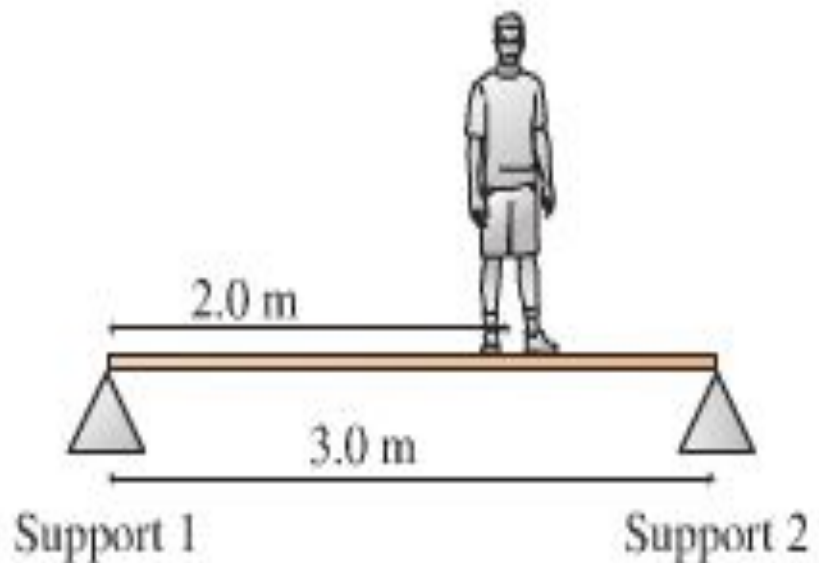
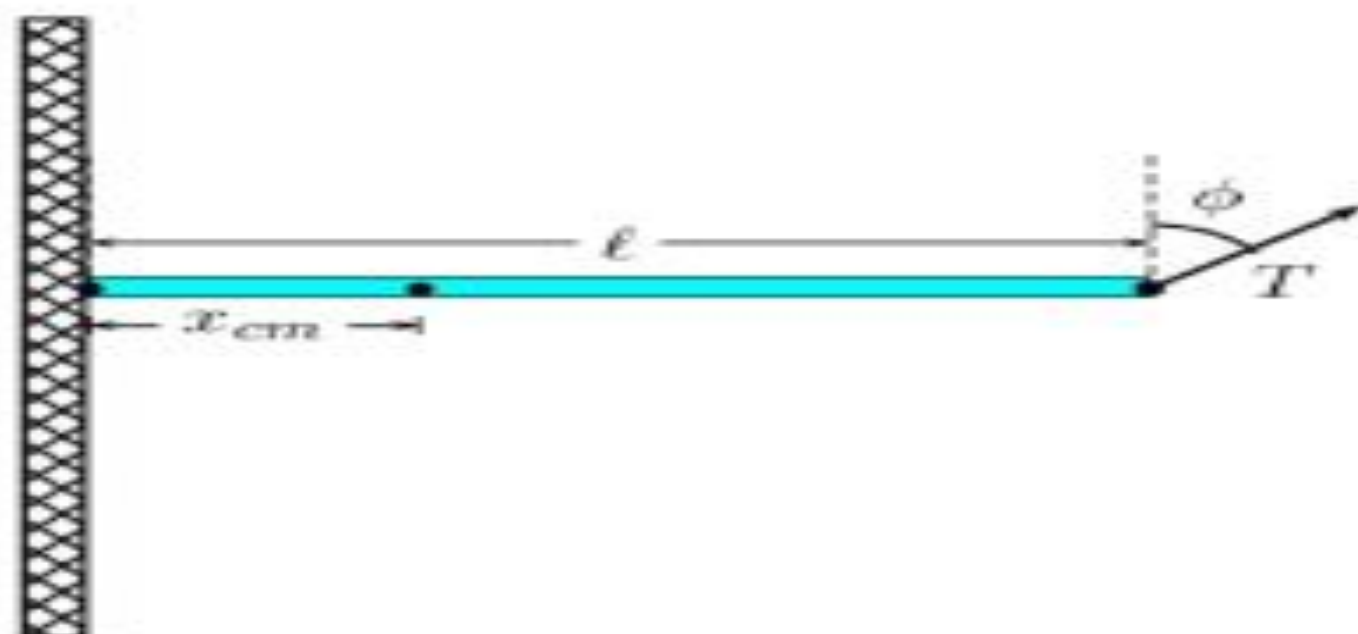


FIGURE P8.43

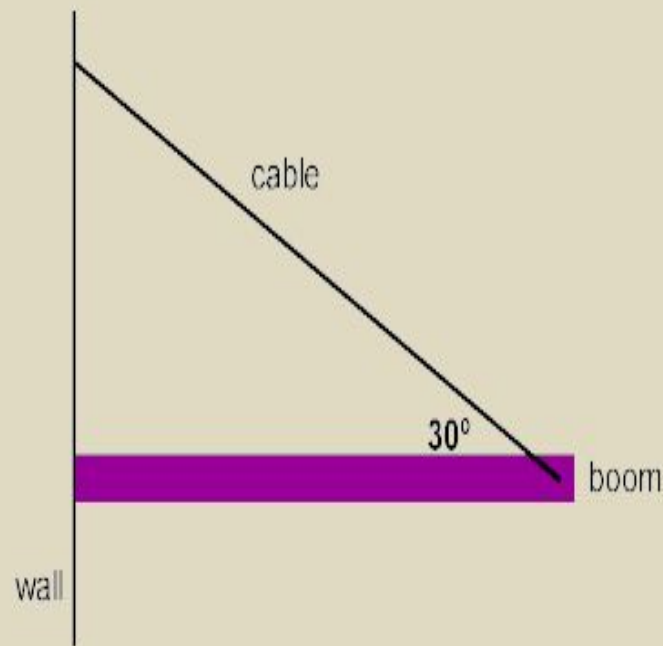
A horizontal, nonuniform beam of mass M and length ℓ is hinged to a vertical wall at one side, and attached to a wire on the other end. The bar is motionless and a wire exerts a force T at an angle of ϕ with respect to the vertical.



If the mass of the beam is 10 kg, the tension in the wire is 50 N, $\sin \phi = \frac{3}{5}$ and $\cos \phi = \frac{4}{5}$, how far is the center of mass of the beam from the hinge? The acceleration due to gravity is 10 m/s^2 .

EXAMPLE – If the uniform boom has a mass of 10 kg, calculate the tension in the cable and calculate the horizontal and vertical forces on the boom from the wall.

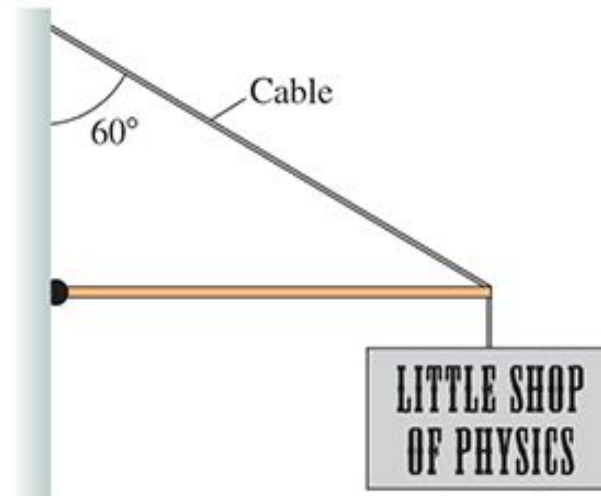
Suppose we have a uniform, horizontal beam and we are asked to calculate the tension in its cable.



BOOM PROBLEM

8. | A vendor hangs an 8.0 kg sign in front of his shop with a cable held away from the building by a lightweight pole. The pole is free to pivot about the end where it touches the wall, as shown in **Figure P8.8**. What is the tension in the cable?

Figure P8.8

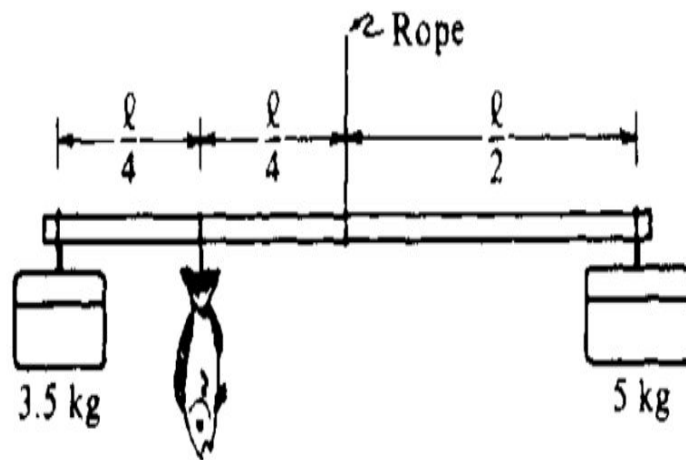


Question:

A turntable begins to rotate from rest with a constant angular acceleration. If a point on the edge of the turntable travels 9 radians during a 3.0 second time period, the angular acceleration of the turntable is

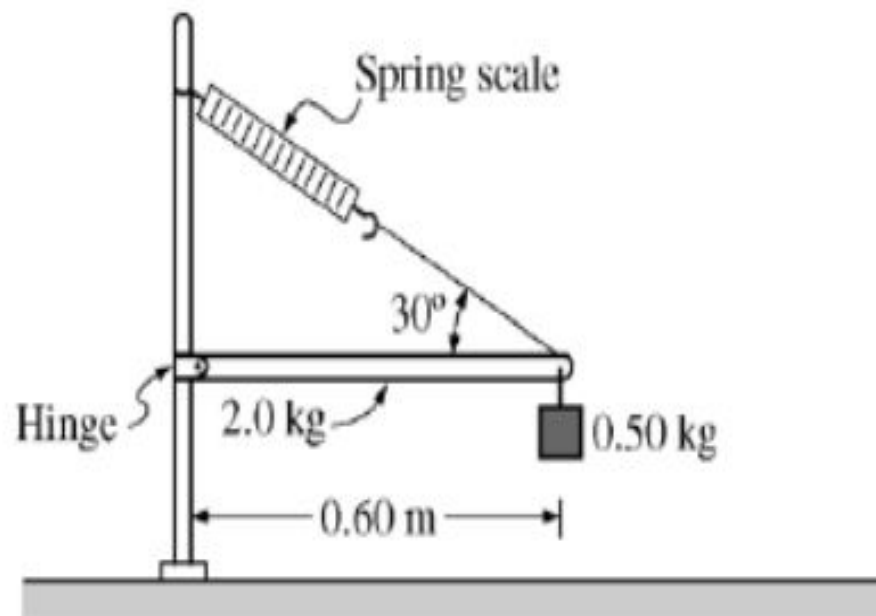
- a. $2.0s^{-2}$
- b. $3.0s^{-2}$
- c. $4.0s^{-2}$
- d. $6.0s^{-2}$
- e. $9.0s^{-2}$

- A uniform meterstick is supported by a string placed at the 40 cm mark. A 50 g mass is placed at the 80 cm mark and a 100 g mass is placed at the 0 cm mark. What is the mass of the meterstick?



_____ 8. To weigh a fish, a person hangs a tackle box of mass 3.5 kilograms and a cooler of mass 5 kilograms from the ends of a uniform rigid pole that is suspended by a rope attached to its center. The system balances when the fish hangs at a point $\frac{1}{4}$ of the rod's length from the tackle box. What is the mass of the fish?

- (A) 2 kg (B) 6.5 kg (C) 1.5 kg (D) 6 kg (E) 3 kg



The horizontal uniform rod shown above has length 0.60 m and mass 2.0 kg. The left end of the rod is attached to a vertical support by a frictionless hinge that allows the rod to swing up or down. The right end of the rod is supported by a cord that makes an angle of 30° with the rod. A spring scale of negligible mass measures the tension in the cord. A 0.50 kg block is also attached to the right end of the rod.

(a) On the diagram below, draw and label vectors to represent all the forces acting on the rod. Show each force vector originating at its point of application.



(b) Calculate the reading on the spring scale.

(c) Calculate the magnitude of the force exerted by the hinge on the rod

•21 ILW The system in Fig. 12-38 is in equilibrium. A concrete block of mass 225 kg hangs from the end of the uniform strut of mass 45.0 kg. A cable runs from the ground, over the top of the strut, and down to the block, holding the block in place. For angles $\phi = 30.0^\circ$ and $\theta = 45.0^\circ$, find (a) the tension T in the cable and the (b) horizontal and (c) vertical components of the force on the strut from the hinge.

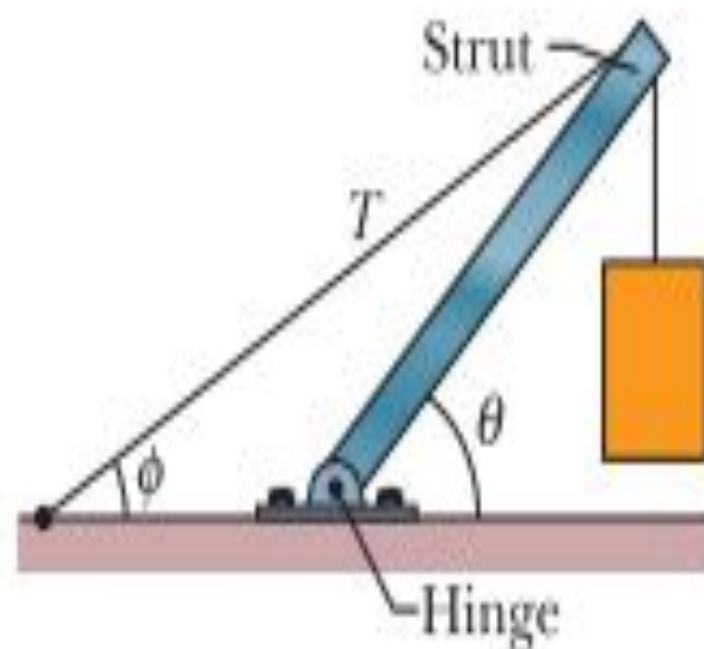


Figure 12-38 Problem 21.

23  In Fig. 12-40, one end of a uniform beam of weight 222 N is hinged to a wall; the other end is supported by a wire that makes angles $\theta = 30.0^\circ$ with both wall and beam. Find (a) the tension in the wire and the (b) horizontal and (c) vertical components of the force of the hinge on the beam.

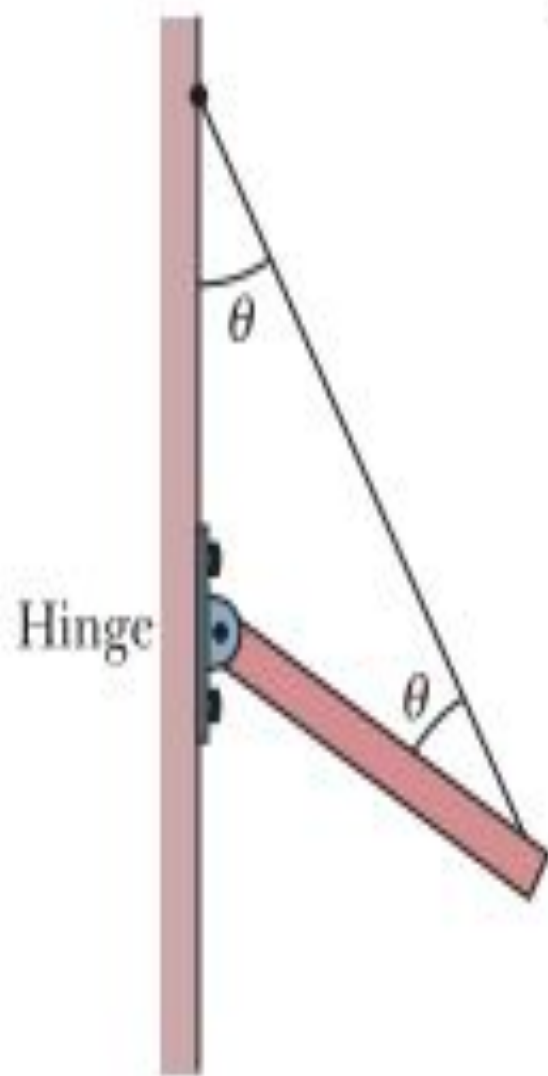


Figure 12-40 Problem 23.